

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

1st September 2021

Subject SA1 – Health and Care

Time allowed: 3 Hours 30 Minutes (14.30 - 18.00 Hours)

Total Marks: 100

INSTRUCTIONS TO THE CANDIDATES

1. *Please read the instructions to examinees sent along with hall ticket carefully and follow without exception.*
2. *The answers are expected to be India Specific application for the syllabus and corresponding core reading. However, substantially the core reading material is still taken from material supplied by Actuarial Education Company which is meant for UK Fellowship examination. The core reading also contains some material which is India Specific, mostly the IRDA regulation. In view of this, it should be noted that focal point of answers is expected to be India Specific application. However, if application specific to any other country is quoted in the answer the candidate should answer the question with reference to Indian environment.*
3. *Mark allocations are shown in brackets.*

Q .1) ‘All is well’ is a medium-sized stand-alone health insurance company offering the following products

- Long-term guaranteed premium Individual income protection insurance offering regular income for periods of total disability
- One-year reviewable Group medical insurance offering reimbursement of medical expenses

i) List the risk factors that drive the claims outgo for each product and the rating factors (practical proxies to the risk factors) that are applied in determining premium rates. (6)

ii) Outline the risks that the company is exposed to in the event of a pandemic. (4)

The company has no reinsurance arrangement currently in place. Concerned about the impact of a potential pandemic, the company is exploring risk transfer arrangements.

iii) Assess the suitability of the following reinsurance arrangements to mitigate the company’s catastrophic risk exposure.

- a) Quota-share reinsurance
- b) Individual Surplus reinsurance
- c) Per person excess of loss reinsurance
- d) Per occurrence excess of loss reinsurance
- e) Aggregate excess of loss reinsurance (15)

iv) Suggest alternative (to reinsurance) risk transfer arrangements to mitigate the company’s catastrophic risk exposure. (5)

The regulator is contemplating the merits of setting up a national catastrophic pool to help mitigate the impact of a pandemic on health insurance companies.

v) Suggest an operating mechanism for such a pool including the factors that should be considered in determining each insurer’s contribution into the pool. (10)
[40]

Q .2) You work in a moderate size health insurance company which sells long term individual Income protection (IP), Critical Illness (CI) and impaired life annuities.

i) Explain briefly how the future investment income from the existing assets could be estimated. (8)

ii) The company is planning to move to Solvency II framework where the Market consistent approach is adopted to place values to its assets & liabilities. Your liabilities are reasonably matched with your existing assets as the company follows the strategy of reasonable matching on an ongoing basis. The Finance Director of the company has expressed her concern about the possible P&L volatility due to the market consistent approach stipulated in Solvency II framework.

Prepare a brief note explaining the valuation principles & methods under Solvency II for:

a) Assets (8)

- b) Liabilities** (15)
 - c) The specific provisions to address her concern on P&L volatility** (2)
- [33]**

- Q .3)** You are an Actuary for a large health insurer in India offering yearly reviewable individual and group health insurance products covering both in-patient and out-patient medical expenses. You have been asked to design a top-up product that allows the existing policyholders an option to increase the Annual Maximum Limit (AML). The AML is the maximum claim amount payable per person per year.
- i) Suggest likely reasons for the top-up product.** (4)
 - ii) Explain how you would price the top-up product.** (7)
 - iii) Explain the types of reserves required to be held for the top-up product.** (5)
 - iv) Suggest controls that could help minimize possible anti-selection risk under the product.** (5)
 - v) The company's marketing director is concerned that the product could lead to cannibalization of the existing products. Comment on the concern.** (6)
- [27]**
